

Technical Delivery Summary (Milestone 3)

Milestone 3 delivery was completed across six core objectives, with all outputs now published in the AmnestyDAO-Token GitHub repository (main branch).

1. Project Handover

The full MVP codebase was consolidated and published, including frontend, backend, Hasura configuration, and Docker environment. Build, run, and deployment documentation has been provided to support reproducibility and future development.

2. Pilot Closure and System Documentation

Following completion of the pilot testing phase, app distribution via Apple App Store and Google Play was formally closed. System architecture and distribution documentation have been updated to reflect the current state of the MVP.

3. Functional Validation

A comprehensive functional validation process was conducted, including:

- Documentation of 21 core user and admin interaction flows
- End-to-end testing suite using Playwright covering full user journeys with live Cardano testnet transactions
- Functional validation report mapping test coverage, execution results, and identified defects with remediation steps

4. Load Testing and Performance Validation

Load testing was conducted using a k6-based framework, validating system performance under concurrent usage. Testing identified a UTxO contention constraint in the reward distribution system, which was addressed through a round-robin optimisation, resulting in a significant improvement in transaction throughput. Full performance metrics and scaling considerations are documented in the repository.

5. Baseline Security Review

A structured security review was completed, including:

- Dependency vulnerability analysis and remediation of critical issues
- Role-based access and permission validation
- Security findings register with prioritised remediation roadmap
- Blockchain implementation review covering validator design and transaction handling

6. System and Token Documentation

Comprehensive documentation has been produced covering system architecture, token flows, technical limitations, and operational considerations, forming a complete technical handover for future development phases.

All testing artefacts, reports, and documentation are publicly available in the repository and form part of the Milestone 3 validation evidence.

Testing was conducted on the Cardano preprod testnet. As protocol parameters evolve, some test conditions may require revalidation in future phases, which has been documented in the technical reports.